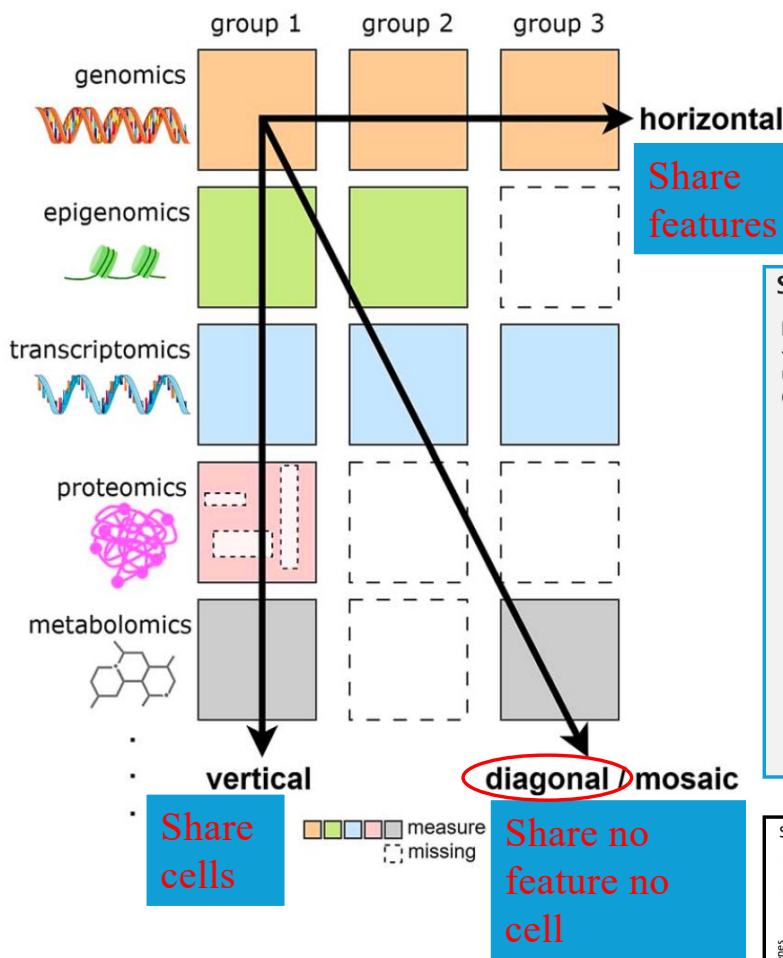




Pure diagonal integration:

- * Share no feature no cell
- * **Topological alignment**
- * Assume similar/shared celltype structures
- * May perform poorly
- * Make more sense for spatial data (PASTE2, STalign)
- * e.g. MMD-MA, PAMONA, SCOT, UniCom.

SCOT works in three steps:

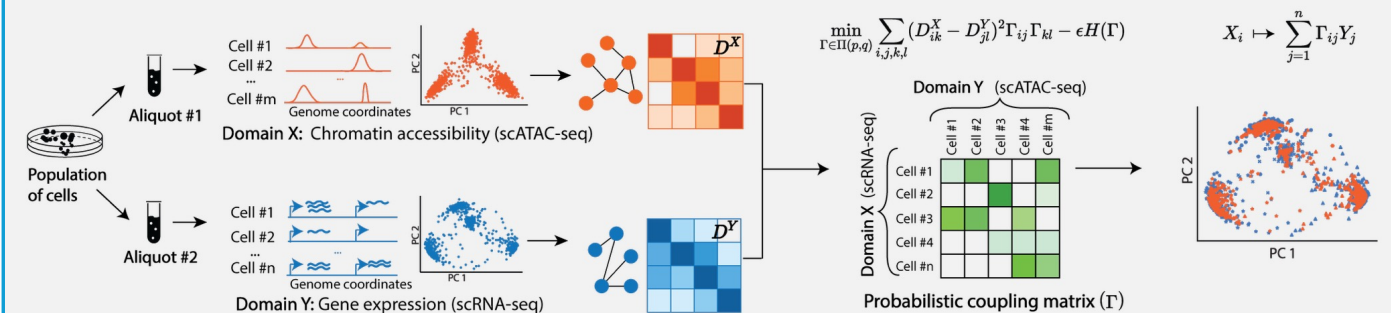
INPUT: Two different single-cell -omics datasets that share some underlying biological manifold (e.g. from the same population)

STEP 1. Computing intra-domain distance matrices by constructing a kNN graph per domain based on correlations and computing shortest distances

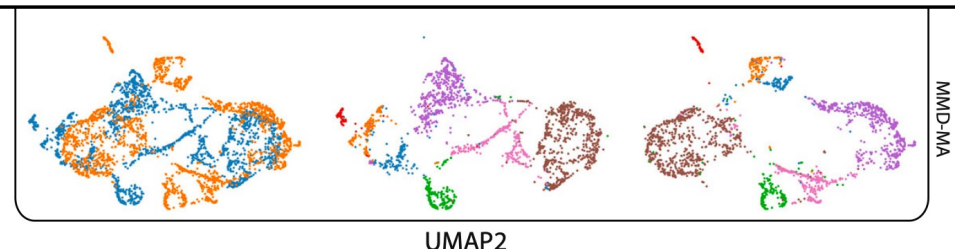
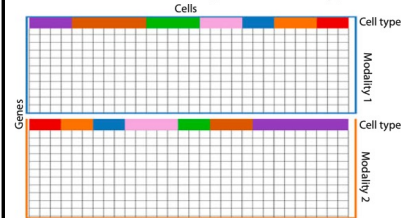
STEP 2. Finding cell-to-cell correspondence probabilities via Gromov-Wasserstein optimal transport with entropic regularization, given D^X , D^Y , and marginal probabilities p , q (uniform distributions defined over the data points in each domain)

STEP 3: Alignment via barycentric projection

OUTPUT: Aligned datasets



Scenario 5: Random sampling from each cell-type



Prior knowledge guidance: Mosaic integration

- **Linked features** (partially share features – horizontal integration)
 - ❖ RNA+ATAC (strong)
 - ❖ RNA+surfaceProtein (weak)
 - **Use only linked features:**
 - E.g. **Seurat_CCA/rPCA**, **LIGER-iNMF**
 - **Linked+unlinked features:**
 - E.g. **scMODEL**, **GLUE** ,
 - **scVI**, **scANVI**, **MultiVI**, **LIGER-UINMF**
- **Leverage joint-profiling** (partially share cells – vertical integration)
 - E.g. **MIDAS**, **Seurat-bridge** , **Stabmap**,
 - **MultiVI**, **Cobolt**, **Multigrade**, **scMoMAT**, **scVAEit**
- **Leverage similar histology in spatial multi-omics**
 - **spatialEx+**

Linear
NMF
Autoencoder

Linked features

scRNA+scATAC

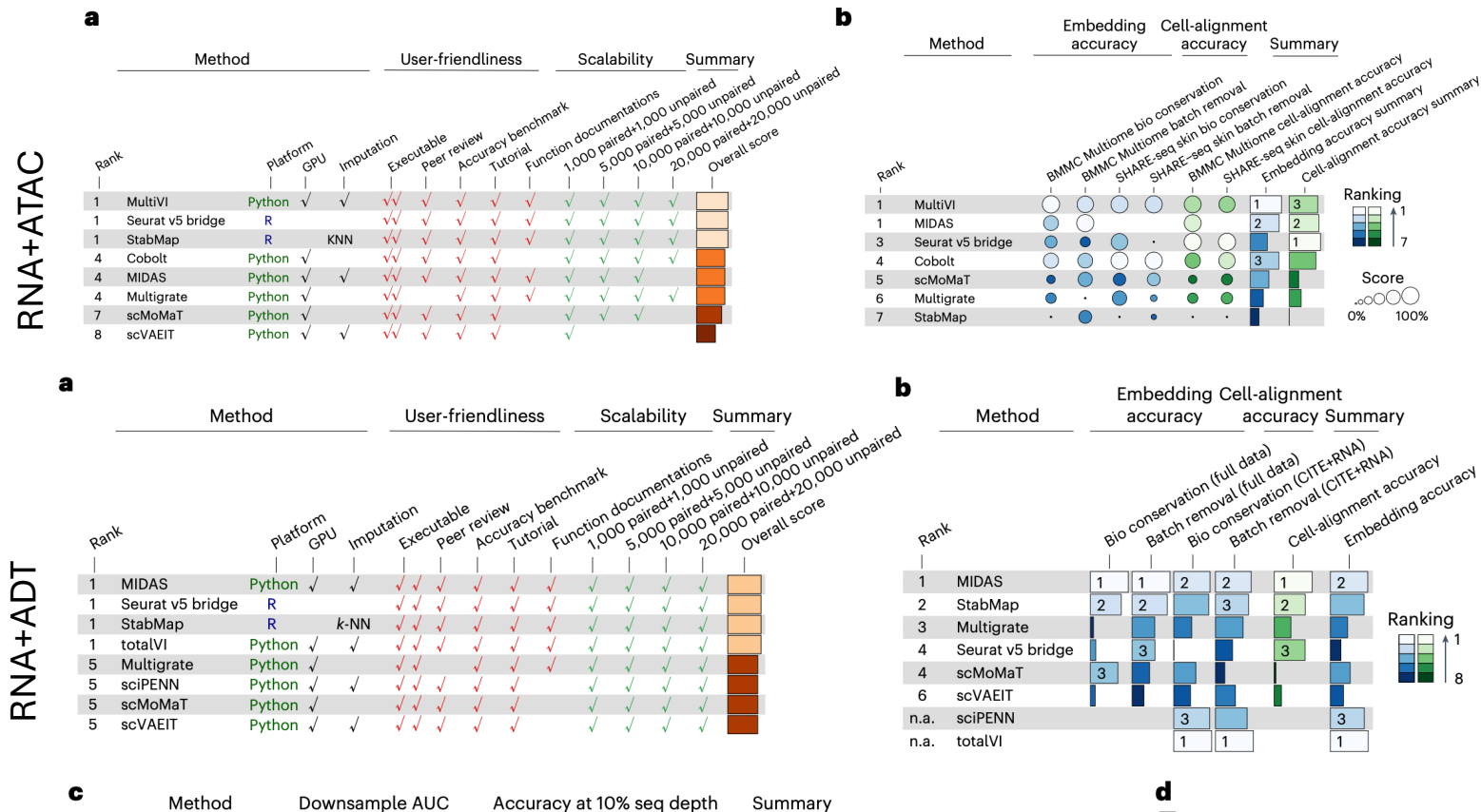
- scATAC summarized to gene activity
- Regulatory interactions: feature correspondence

a

Method			User-friendliness					Scalability						Summary		
Rank		Platform	GPU	Executable	Peer review	Accuracy benchmark	Tutorial	Function documentations	500	1,000	5,000	10,000	50,000	100,000	500,000	Overall score
1	GLUE	Python	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	LIGER iNMF	R		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	LIGER online iNMF	R		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	LIGER UINMF	R		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	MaxFuse	Python		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	SCALEX	Python	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	Seurat v3 CCA	R		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	SIMBA	Python		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	uniPort	Python	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
10	bindSC	R		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
10	MultiMAP	Python		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
12	Pamona	Python		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
13	unionCom	Python	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
14	GCN-SC	Python	✓	✓												

Fu et al. 2025 Nature methods. <https://doi.org/10.1038/s41592-025-02737-9>

Leverage joint-profiling



Fu et al. 2025 Nature methods. <https://doi.org/10.1038/s41592-025-02737-9>

Feature	MultiVI	Cobolt
Modality support	RNA + ATAC (extendable to more)	RNA + ATAC + protein (flexible multimodal)
Handles mosaic cells?	✅ Yes (cells can have missing modalities)	✅ Yes (cells can have missing modalities)
Model type	VAE (modality-specific encoders → joint latent)	VAE (modality-specific encoders → joint latent)
Latent alignment	KL + prior regularization	KL + adversarial alignment + contrastive learning
Use case	Multiome, scRNA+ATAC, integration across batches	Flexible multimodal integration, can combine very heterogeneous modalities

◆ 3 Cobolt Details

- **Architecture:**
 - Modality-specific encoders → latent embeddings
 - Alignment is done using **contrastive loss** or **adversarial methods**
 - Decoder reconstructs observed modalities
 - Can flexibly handle **any combination of RNA, ATAC, protein, etc.**
- **Strengths:**
 - Very flexible for **heterogeneous multimodal datasets**
 - Better alignment when modalities are very different
 - Can explicitly enforce **latent space modality alignment**
- **Limitations:**
 - Slightly more complex to train
 - Requires careful tuning of **contrastive/adversarial losses**
 - May scale slower than MultiVI for very large datasets

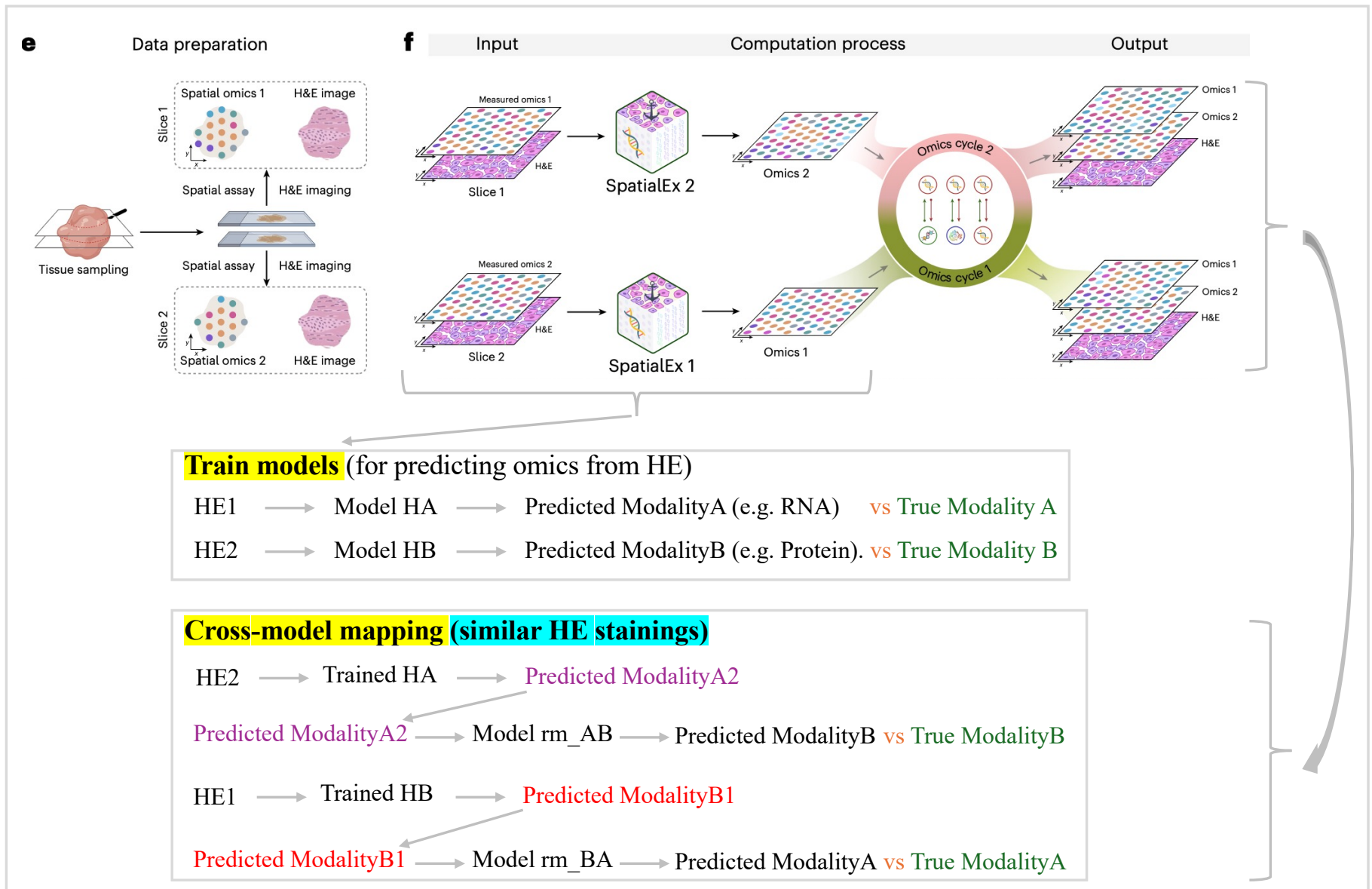
1. MultiVI:

- VAE + KL alignment
- Handles missing modalities naturally
- Focus on **probabilistic reconstruction**

2. Cobolt:

- VAE + contrastive/adversarial alignment
- Designed for **stronger alignment across highly heterogeneous modalities**
- Also handles missing modalities (mosaic cells)

Intuition: MultiVI is simpler, probabilistic, and scales well; Cobolt is more flexible and can produce tighter latent alignment across very different modalities.



Feature	scVI	totalVI	MultiVI
Primary use	Single-cell RNA-seq (scRNA) integration & batch correction	RNA + protein (CITE-seq) integration	RNA + ATAC integration (scRNA + scATAC)
Modalities supported	RNA only	RNA + protein	RNA + chromatin accessibility
Paired cells required?	No (for batch correction)	Yes (RNA + protein measured in same cell)	Optional (can integrate unpaired RNA & ATAC datasets)
Output	Latent space embeddings	Shared latent space embeddings & imputed protein expression	Shared latent space embeddings & imputed accessibility / RNA
Batch correction	Yes	Yes	Yes
Label transfer / annotation	scANVI extension	Can transfer labels using latent space	Can transfer labels using latent space
Imputation	RNA counts	RNA + protein	RNA + ATAC gene activity
Typical applications	Batch correction, clustering, trajectory inference	CITE-seq analysis, protein prediction, clustering	Joint scRNA + scATAC analysis, multi-omics clustering, label transfer

Pure diagonal integration:

Topological alignment

Assume the same/shared celltype structures

Make more sense for spatial data

MMD-MA

1. Each dataset: low dimensional representation → KNN graph → distance metrics
2. Aligns the cells across single-cell multi-omics datasets by matching the distance matrices via minimizing the **maximum mean discrepancy** between the two datasets.

UnionCom

1. Each dataset: low dimensional representation → KNN graph
→ distance metrics
2. Aligns the cells across single-cell multi-omics datasets by matching the distance matrices via **a matrix optimization method**

SCOT

SCOT works in three steps:

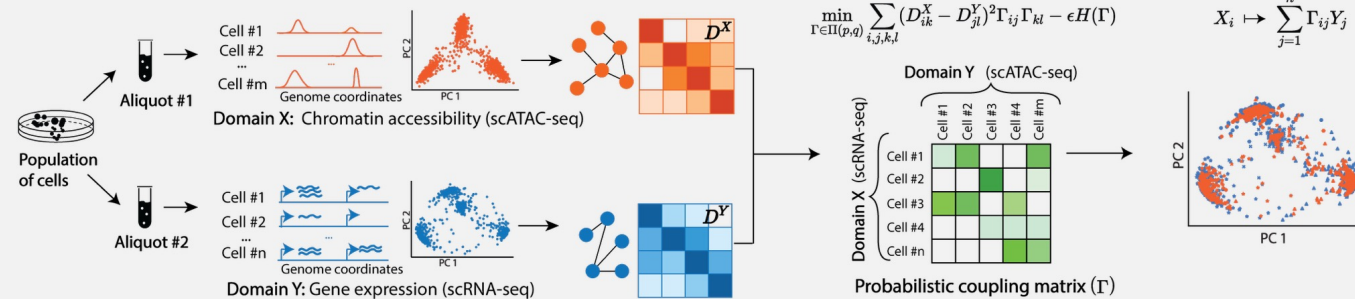
INPUT: Two different single-cell -omics datasets that share some underlying biological manifold (e.g. from the same population)

STEP 1. Computing intra-domain distance matrices by constructing a kNN graph per domain based on correlations and computing shortest distances

STEP 2. Finding cell-to-cell correspondence probabilities via Gromov-Wasserstein optimal transport with entropic regularization, given D^X , D^Y , and marginal probabilities p , q (uniform distributions defined over the data points in each domain)

STEP 3: Alignment via barycentric projection

OUTPUT: Aligned datasets



<https://rsinghlab.github.io/SCOT/>

1. Both datasets → low dimensional representation → distance matrices
2. Cross-modality cell alignment (**GW**, align geometry, representing celltype structures)

Partial Gromov-Wasserstein alignment

1. Align distance metrics (representing graph/geometry)
2. Find cross-dataset cell-cell correspondence that best preserve geometry in each dataset.
3. Look for corresponding cells that are similarly positioned relative to its neighborhood in the two datasets
4. **Require similar cell-type structure in both datasets (e.g. from the same tissue)**
5. Make more sense when aligning spatial data, e.g. in PASTE2, STalign

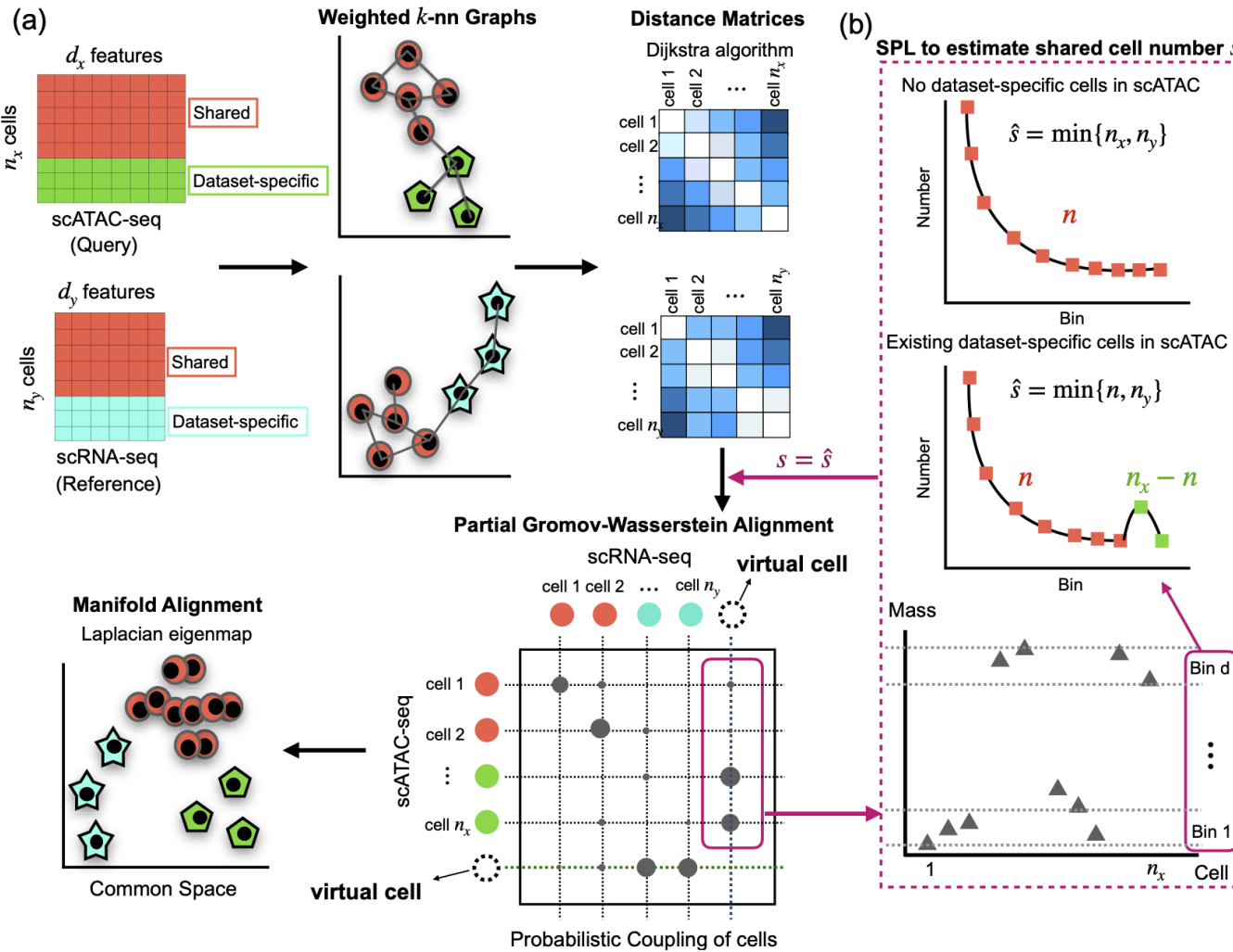


Fig. 1. Overview of Pamona. Pamona is a partial manifold alignment algorithm for heterogeneous single-cell multi-omics data integration. Given inputs of multiple cellular modalities (e.g. scATAC-seq and scRNA-seq), it identifies both shared and dataset-specific cells based on the computed probabilistic couplings of cells across datasets, and it aligns cellular modalities in a common low-dimensional space, while simultaneously preserving both shared and dataset-specific structures. (a) Pamona constructs a weighted k -nn graph of cells for each dataset (step 1), computes the geodesic distance matrix of cells within each dataset (step 2), computes the probabilistic coupling matrices of cells based on the partial Gromov-Wasserstein optimal transport (step 3), and aligns cellular modalities with distinct unmatched features in a common low-dimensional space to holistically represent the cellular structures (step 4). (b) A Scree-Plot-Like (SPL) method is proposed to estimate the shared cell number s when it is not available

1. Both datasets $\rightarrow$ low dimensional representation $\rightarrow$ distance matrices
2. Cross-modality cell alignment (**partial** GW, align geometry, representing celltype structures)

Partial Gromov-Wasserstein alignment

1. Align distance metrics (representing graph/geometry)
2. Find cross-dataset cell-cell correspondence that best preserve geometry in each dataset.
3. Look for corresponding cells that are similarly positioned relative to its neighborhood in the two datasets
4. Require **overlapping** cell-type structure in both datasets
5. Make more sense when aligning spatial data, e.g. in PASTE2, STalign

NMF (non-negative matrix factorization)

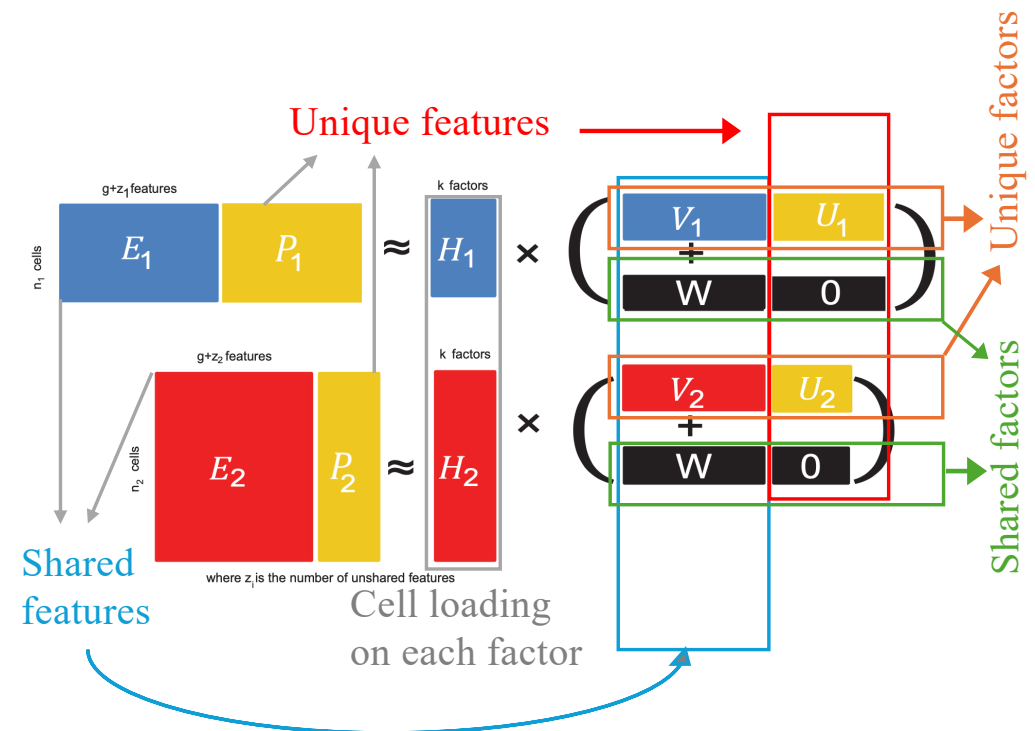
All feature loading/weights on each factor: positive, only positively correlated features

LIGER: iNMF

- Embedde cells together in a shared low dimensional space through iNMF
- Incorporate only shared features (for cell correspondence)

LIGER: UINMF

- Embedde cells together in a shared low dimensional space through UINMF
- Incorporate both shared (for cell correspondence) and unique features (more information)



Kriebel&Welch 2022 Nature communication

Adversarial alignment

Learn a representation where cells from different datasets cannot be distinguished.

The Architecture

Adversarial alignment has two components:

Encoder (Feature extractor)

Maps expression into latent space:

$$Z = f(X)$$

Discriminator

Tries to predict:

- Which dataset did this cell come from?

$$D(Z) \rightarrow dataset_label$$

The Adversarial Game

We train them against each other:

- Discriminator tries to detect slide A vs slide B
- Encoder tries to fool the discriminator

So the encoder learns:

A representation where slide information disappears.

What This Achieves

You get:

- Shared embedding space
- Slide-invariant cell representations
- Biological similarity preserved
- Technical differences minimized

GAN: generative adversarial networks

	GAN	Adversarial Alignment
Goal	Generate realistic data	Align embeddings
Fake data produced?	Yes	No
Generator?	Yes	No (encoder only)
Discriminator?	Real vs fake	Dataset A vs B
Used for	Simulation	Integration

1 Normal Encoder (Deterministic)

Example:

Python

```
z = Encoder(x)
```



- **Type:** Deterministic mapping
- **Output:** A single latent vector per cell
- **Loss:** Usually reconstruction loss (MSE, cross-entropy, etc.)
- **Batch effect:**
 - The encoder **will preserve any systematic differences** in input.
 - No batch correction happens **unless you explicitly enforce it** (contrastive loss, adversarial, etc.).
- **Pros:** Simple, fast
- **Cons:** Cannot automatically model uncertainty or account for batch effects

2 scVI-style Encoder (Variational)

Encoder:

$$q(z \mid x, batch)$$

- **Type:** Probabilistic / stochastic
- **Output:** Latent **distribution** (mean μ , variance σ^2)
- **Loss:** Variational ELBO

$$\mathcal{L} = \text{Reconstruction Loss} + \text{KL divergence}$$

- **Batch modeling:** Batch is passed as an **explicit covariate** to the decoder
 - Decoder reconstructs x conditional on batch: $p(x \mid z, batch)$
 - Latent z learns to capture **biological signal only**, not technical batch variation
- **Batch correction happens because:**
 - The decoder knows batch
 - The ELBO forces latent z to explain only what is common across batches
 - Encoder adapts to produce a batch-invariant z
- **Pros:**
 - Handles batch effects automatically
 - Models uncertainty
 - Can generate new samples
- **Cons:**
 - More complex, slower to train



VAE, ELBO

2 Variational Autoencoder (VAE)

A VAE introduces probability.

Instead of encoding a cell into a single vector z , we encode it into a *distribution*:

$$q(z|x) = \mathcal{N}(\mu(x), \sigma(x))$$

Now:

- Each cell is represented by a Gaussian
- We sample z from that Gaussian
- The decoder reconstructs x

This makes the model **generative**.

3 What is ELBO?

ELBO = Evidence Lower Bound

It is the training objective of a VAE.

In probabilistic modeling, we want to maximize:

$$\log p(x)$$

But this is usually intractable.

So we instead maximize a *lower bound* of it:

$$\text{ELBO} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - KL(q(z|x)||p(z))$$

That's the VAE loss.

4 Understanding the Two Terms

1 Reconstruction Term

$$\mathbb{E}_{q(z|x)}[\log p(x|z)]$$

- Makes reconstructed gene expression match input
- Similar to autoencoder reconstruction loss

In scRNA/spatial models:

- Usually negative binomial likelihood
- Not simple MSE

2 KL Divergence Term

$$KL(q(z|x)||p(z))$$

- Forces latent distribution to resemble a prior
- Usually standard normal:

$$p(z) = \mathcal{N}(0, I)$$

This:

- Smooths latent space
- Prevents overfitting
- Makes it generative



5 Why It's Called "Lower Bound"

We want:

$$\log p(x)$$

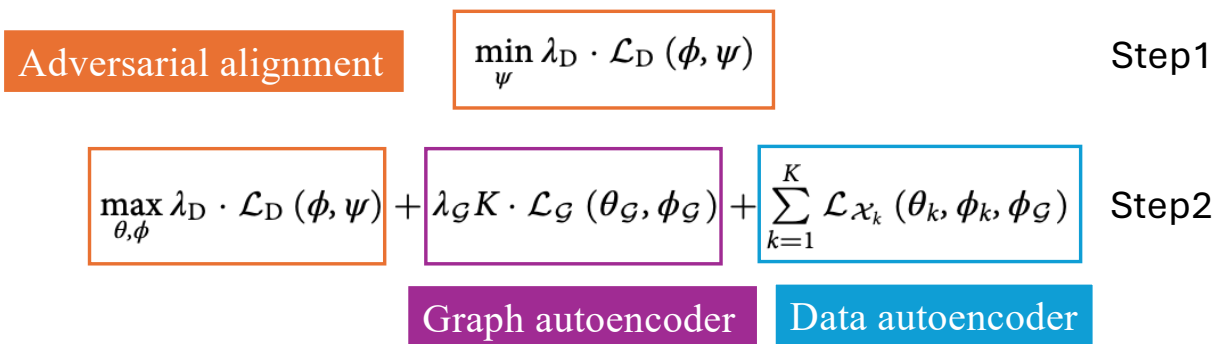
ELBO is mathematically proven to satisfy:

$$\text{ELBO} \leq \log p(x)$$

So maximizing ELBO $\approx$ maximizing data likelihood.

GLUE

- Autoencoder framework
- **Data autoencoder: Align cells**
 - Modality-specific variational autoencoders (**unit: cells**)
 - Use both linked & unlinked features in each modality
 - Handle batches by modeling it
 - **Weighted adversarial alignment**: align cells of different modalities to the same low-dimensional space
- **Graph autoencoder : Link features**
 - Knowledge -based graph (**unit: features**):
 - Knowledge: By default, an ATAC peak is connected to a gene if they overlap in either the gene body or promoter region.
 - **link features from different modalities**
- Model optimization:
 - Optimize Adversarial alignment and Model probability (ELBO)
- **Limit: not work well on weakly linked features, like RNA+surface Proteins**



scMODEL: Linked features, even weakly linked features

- Integrate multi-omics datasets with limited shared features no sharing samples, e.g. RNA+ATAC+Protein
 - RNA-ATAC: strong connection
 - RNA-surface Protein: weak relationship (for guiding integration)
- Deep learning based, autoencoder framework
 - Modality-specific latent variable (encoder)
 - Apply generative adversarial networks to align cell embeddings
 - Guided by MNN: used linked features to look for MNNs as anchors
 - Preserve geometric structure/biological information of each modality
- Limitation: batches
- GPU, large dataset

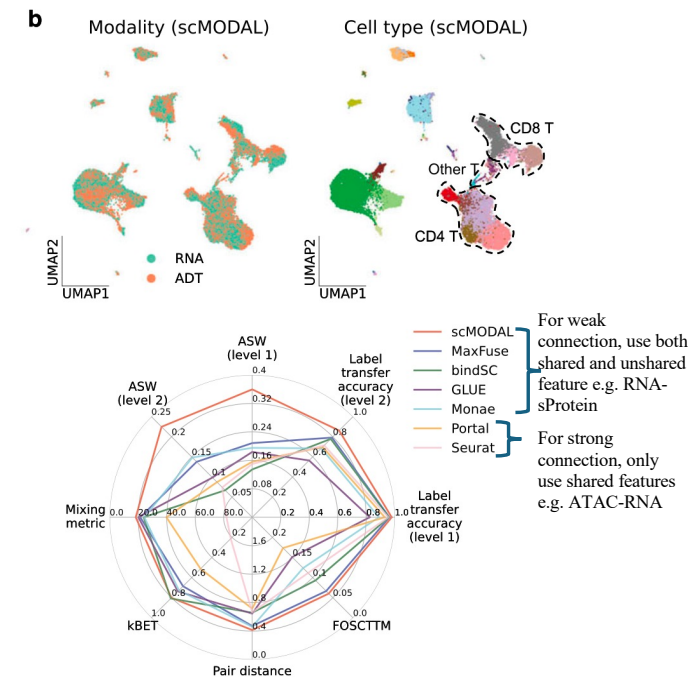


Fig. 1 | Overview of scMODEL. **a** scMODEL takes single-cell feature matrices from different modalities, together with feature links as input. **b** scMODEL utilizes generative adversarial learning to mix the distributions of cell embeddings from different datasets. To find correct correspondence between modalities as well as preserve biological variation within each modality, regularizations to narrow the distance between anchors based on prior information and preserve geometric

representation of cells are applied in the training process of scMODEL. **c** scMODEL outputs integrated cell representations for further analyses, and the composition of trained networks enables imputation of features and inference of feature relationship across single-cell modalities. The results can also be used for multiple downstream analyses, including label transfer for revealing cell identities and cell-cell communication inference using imputed features.

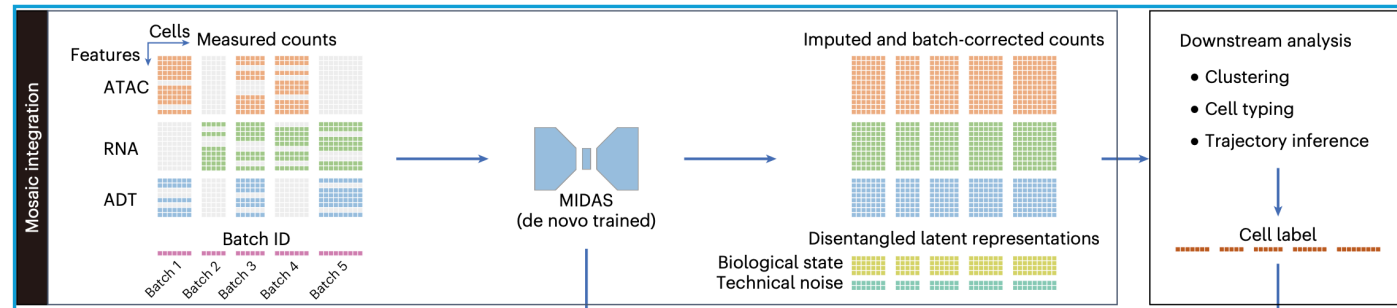
ADT: antibody-derived tag

Feature	scMODEL	Cobolt
Modality support	Flexible: RNA + ATAC + protein, multiome, multi-modal single-cell data	RNA + ATAC + protein, flexible multimodal
Handles mosaic data?	✔ Yes, can handle missing modalities	✔ Yes, can handle missing modalities
Model type	VAE + modality-specific latent variables + optional hierarchical structure	VAE + modality-specific encoders + adversarial/contrastive alignment
Latent alignment	Uses modality-specific latent variables and shared latent space via probabilistic constraints	Uses adversarial or contrastive loss to align latent space across modalities
Decoder	Modality-specific decoders for reconstructing observed data	Modality-specific decoders for observed data
Use case	Joint modeling of multiple modalities, probabilistic inference, can capture modality-specific and shared variation	Strong latent alignment across heterogeneous modalities, flexible multimodal integration

MIDAS:

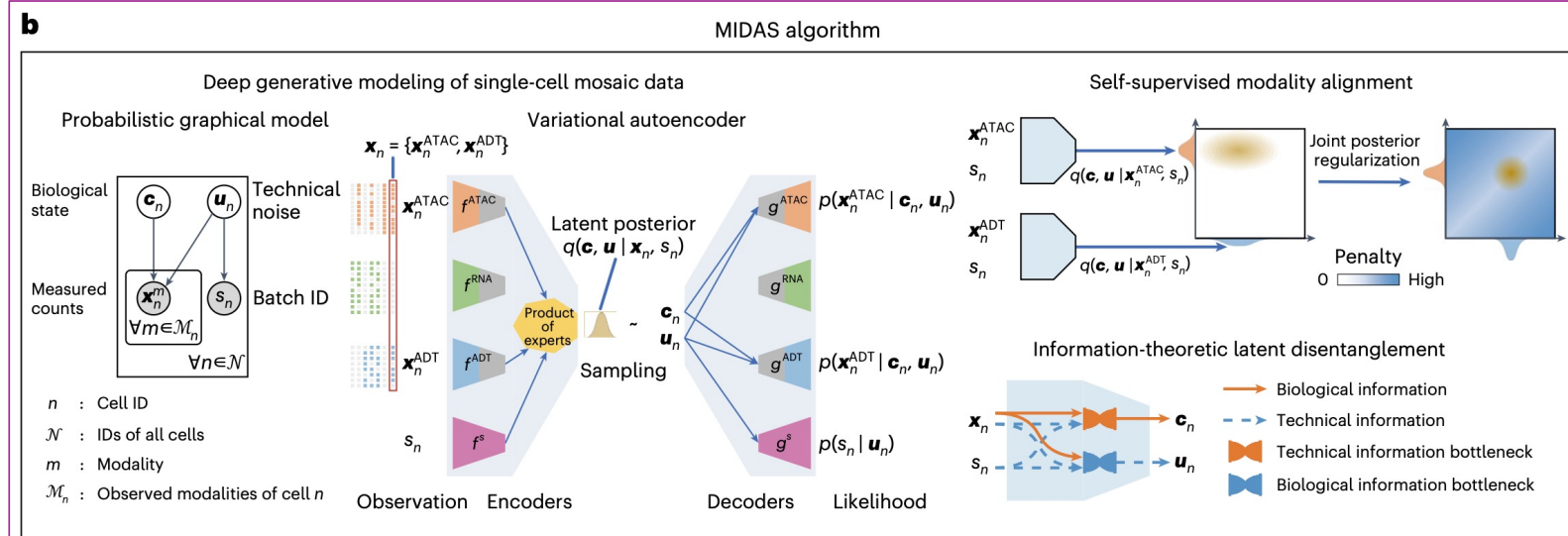
Variational autoencoder that use probabilistic modelling

Designed particularly to leverage joint-profiling of multi omics data



Variational autoencoder

- Indi-modality autoencoder
 - Two sets of latent variables/factors (biological vs batch)
 - Information bottleneck to better separate biological vs batch factors.
- Self-supervised modality alignment: Joint tuning to minimize joint-factor dispersion



scVI

Variational autoencoder that use probabilistic modelling that count for seq depth and batches

- Using both shared and unique features
- Small dataset is not where scVI shines

3 How scVI Works (Pipeline)

1. **Input:** raw counts matrix X (cells $\times$ genes) and optional batch labels
2. **Encoder:** maps $X \rightarrow$ latent z
 - Probabilistic: outputs mean & variance for Gaussian latent variable
3. **Decoder:** maps latent $z \rightarrow$ reconstructed expression $\hat{X}$
 - Models counts via Negative Binomial
4. **ELBO loss:**

Seq depth + batch

$$\text{ELBO} = \underbrace{\mathbb{E}_{q(z|x)}[\log p(x|z)]}_{\text{reconstruction}} - \underbrace{KL(q(z|x)||p(z))}_{\text{latent regularization}}$$

- Maximizing ELBO trains the model
- KL term ensures smooth, compressed latent space
- Reconstruction term ensures biological information is preserved

scANVI: batch correction + cell annotation

1 Relationship Between scVI and scANVI

- **scVI**: learns **latent embeddings** from raw counts, optionally batch-corrected.
- **scANVI**: builds on scVI, adding a **semi-supervised classifier head** to predict labels for unlabeled cells.

Conceptually, scANVI reuses scVI's encoder/decoder weights and adds a new layer for categorical labels.

2 Why this is like Transfer Learning

- Transfer learning = **reuse a pre-trained model for a new task**
- scANVI **reuses the scVI latent space** as a pretrained feature representation
- The new task = **label prediction** for semi-supervised cells
- The classifier head is trained **on top of the fixed or slightly adjusted latent space**

✓ So scANVI uses **transfer learning principles**: leverage knowledge from scVI latent space for a new downstream task (label prediction).

4 Training Objective

The total loss combines:

$$\text{Loss}_{\text{scANVI}} = \underbrace{\text{ELBO}_{\text{scVI}}}_{\text{latent reconstruction}} + \underbrace{\alpha \cdot \text{CrossEntropy}(y, \hat{y})}_{\text{semi-supervised label prediction}}$$

- ELBO = keeps latent space informative and batch-corrected
- CrossEntropy = trains the new categorical head
- If **latent weights are updated**, it's **fine-tuning**

5 Practical Takeaways

Use case	How scANVI behaves
Pretrained scVI weights frozen	Transfer learning: only classifier is trained
Pretrained scVI weights updated	Fine-tuning: latent space + classifier co-adapt
Output	Batch-corrected latent space + predicted categorical labels

```
1 import scvi
2
3 # Setup AnnData
4 scvi.model.SCANVI.setup_anndata(adata, labels_key="cell_type", unlabeled_category="Unknown", batch_key="slide")
5
6 # Initialize model
7 model = scvi.model.SCANVI(adata, unlabeled_category="Unknown")
8
9 # Train model (batch correction + label transfer)
10 model.train(max_epochs=100, use_gpu=True)
11
12 # Get batch-corrected latent embeddings
13 adata.obsm["X_scANVI"] = model.get_latent_representation()
14
15 # Predict labels for unlabeled cells
16 predicted_labels = model.predict(adata)
```

totalVI (vertical integration, RNA+surfaceProtein)

◆ 2 Why totalVI cannot handle mosaic data

1. **Assumption:** totalVI expects **paired measurements for all cells**
 - Every cell has RNA counts **and** protein counts
2. **Model structure:**
 - One encoder → maps RNA + protein into **joint latent space**
 - Decoder → reconstructs both RNA and protein for **all cells**
3. If a cell is **missing protein**, totalVI cannot compute the reconstruction loss for that modality
 - Training breaks because the VAE expects a **complete observation vector**

✅ So totalVI = **full-modal paired data only**

◆ 3 Why multiVI can handle mosaic data

1. **Separate modality encoders:**
 - RNA encoder → latent representation Z^R
 - ATAC encoder → latent representation Z^A
2. **Modality-specific likelihoods:**
 - Each modality has its own decoder
 - The reconstruction loss is computed **only for observed modalities**
 - If RNA is missing → skip RNA reconstruction
 - If ATAC is missing → skip ATAC reconstruction
3. **Joint latent space:**
 - MultiVI aligns modality-specific latent spaces using **KL divergence and prior regularization**
 - Missing modalities do not prevent latent inference
4. **Result:**
 - MultiVI can integrate cells with **any subset of modalities**
 - Ideal for **mosaic datasets** (some cells with RNA only, some with RNA+ATAC, etc.)

✅ Key Point:

multiVI works on mosaic data because it **computes reconstruction loss only for observed modalities**, while still aligning modality-specific latent spaces.
totalVI assumes **all cells have both RNA and protein**, so missing modalities break training.

multiVI

Variational autoencoder that use probabilistic modelling that count for seq depth and batches

- Using both shared and unique features
- Horizontal integration
- Vertical integration
- Mosaic integration (missing modality)
- Data imputation

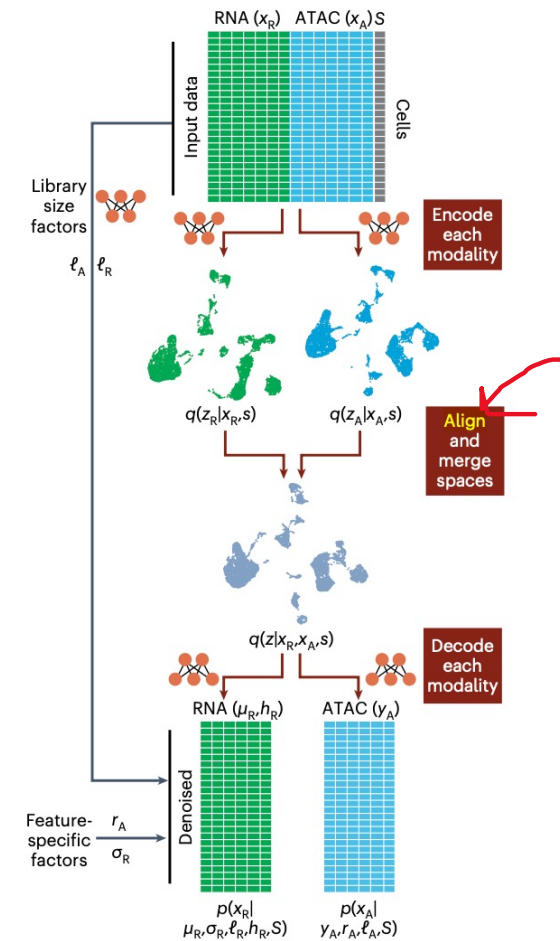
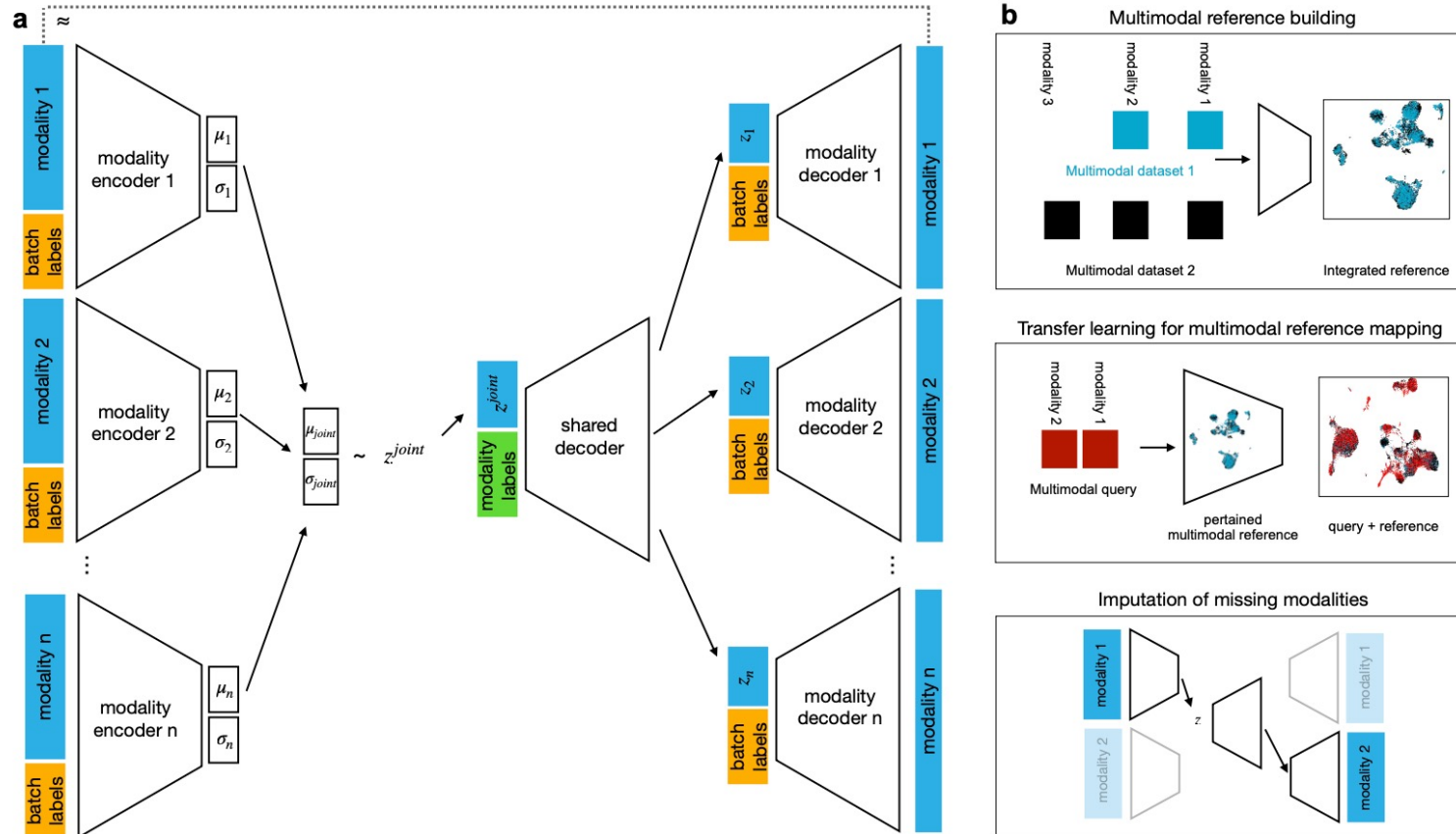


Fig. 1 | Conceptual model illustration in which input data (top) consist of chromatin accessibility (ATAC), gene expression (RNA) or both data types (multiome). Variable S represents experimental covariates, such as batch or experimental condition. Each data modality is encoded into modality-independent latent representations (using neural network encoders) and then, these representations are merged into a joint latent space. The joint latent representation is used to estimate (decode) the input data together with chromatin region-specific effects (r_A), gene-specific dispersion (σ_R), cell-specific effects (ℓ_A, ℓ_R), accessibility probability estimates (y_A) and mean gene expression values (μ_R).

Multigrate: multi-omic data integration



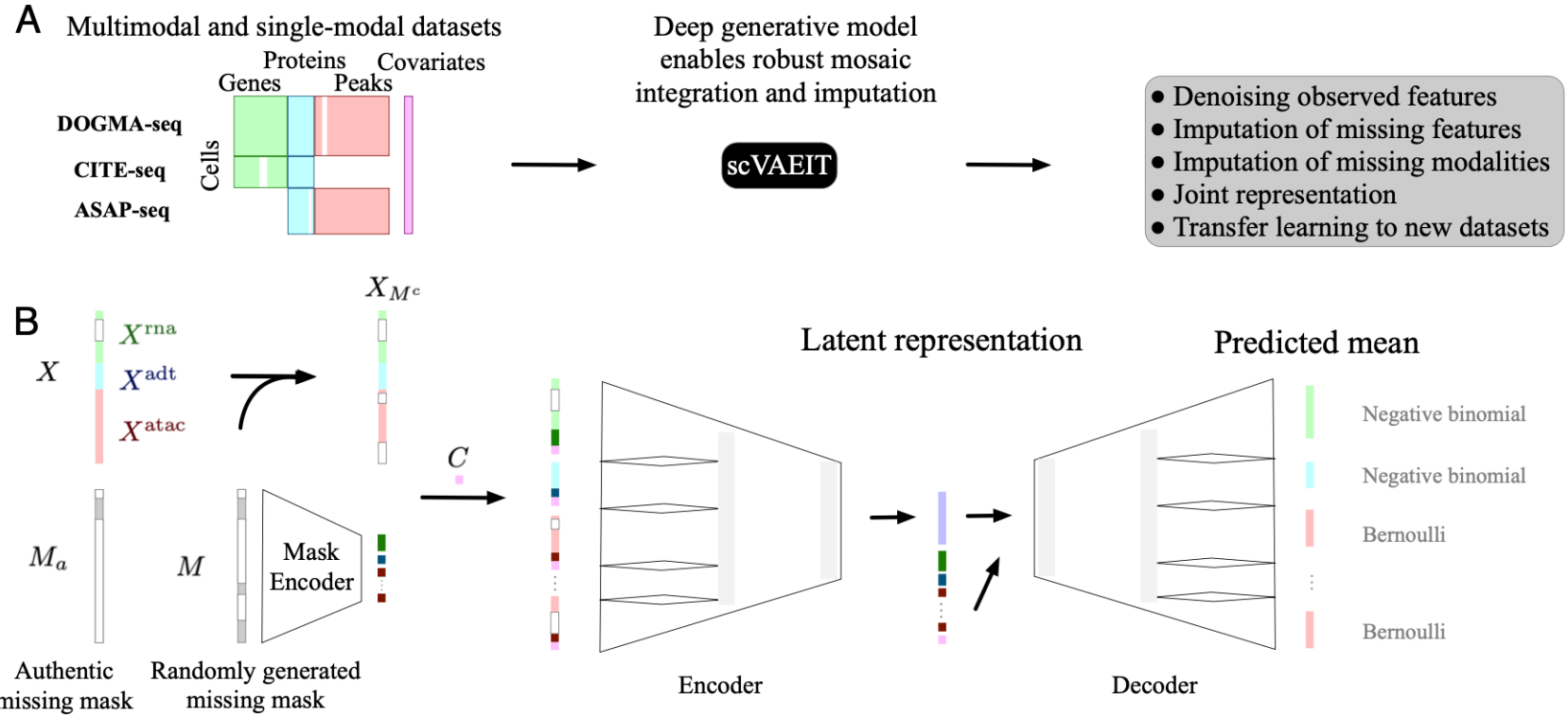


Fig. 1. Overview of multimodal single-cell datasets mosaic integration with scVAEIT. (A) Multimodal sequencing techniques such as DOGMA-seq, CITE-seq, and ASAP-seq simultaneously measure multiple modalities in single cells. Different datasets can measure different panels of features for the same modality, producing mosaic count matrices for RNA, proteins and peaks. scVAEIT takes these matrices and an optional matrix containing continuous and categorical covariates to learn cross-modality and cross-feature relationships. It also produces a joint representation of all modalities for each cell. (B) For each cell, scVAEIT uses a mask to inform the missingness of the data. The actual or authentic mask is M_a . During training, masks (M) are randomly generated and encoded for each modality to force the model to learn to predict specific portions of the features based on other observed features (X_{MC}). In the first layer of the encoder and the last layer of the decoder, different modalities and peaks in different chromosomes break into subconnections that greatly reduces model complexity. The encoder then outputs the parameters of the posterior distributions for the latent variable Z , given the unmasked data X_{MC} and the mask M . Next, samples from the posterior distributions along with the mask and covariates are fed to a decoder neural network to predict the posterior mean of X . Ultimately unobserved values are imputed and observed values are denoised.

cell2location

Aims of integration

- Joint visualization
- Clusterings
- Label transfer

- Purposes of Integration: Visualization, Clustering, Label transfer
- Seurat RNA+ATAC: require shared features? https://satijalab.org/seurat/articles/seurat5_atacseq_integration_vignette

Many deep learning-based methods integrate datasets based only on shared features, such as Seurat, scVI, scANVI, VIPCCA, SCALEX, iMAP and Portal, disregarding valuable unshared features.

Seurat+WNN: shared cells, multi-omics profiling

Recent efforts have incorporated unshared features into single-cell multi-omics integration. Methods such as totalVI, MultiVI, CLUE, MIDAS, scButtery, and SpatialGLUE attempt to integrate multi-omics data by leveraging autoencoders on joint-profiled datasets, but require (partially) paired cells across modalities.

To address cross-modality integration using unpaired single-cell data, alternative approaches leveraging autoencoders and GANs have been proposed. For instance, scMMGAN76 employs GANs to learn multi-modal mappings. However, it integrates datasets in a paired manner and does not provide a shared latent space that captures a unified view across all modalities.

Other methods, including GLUE, CoVEL, and Monae, incorporate prior knowledge of cross-modality interactions—mainly between genes and epigenomic profiles such as ATAC peaks—by linking them in a knowledge-based graph. However, our benchmarking experiments indicate that the graph-variational autoencoder-based approach for incorporating prior knowledge of cross-modality interactions in these methods perform less effectively when leveraging prior interaction information between weakly linked modalities, such as RNA and protein.

Unlike these methods, scMODAL has unique designs for robustly integrating single-cell datasets across modalities with varying feature connections. Instead of relying on knowledge graph-based approaches like those used in GLUE, CoVEL, and Monae to incorporate prior information on feature relationships,

scMODAL leverages linked features from prior knowledge to construct MNN pairs, which serve as potential anchors for GAN alignment. By minimizing the latent distance between these MNN pairs as a regularizer during GAN training, scMODAL enables a more flexible utilization of prior feature link information. This approach reduces dependence on the strength of cross-modality feature links while ensuring effective latent space mixing, enabling scMODAL to not only integrate strongly linked modalities such as RNA and ATAC, but also integrate weakly linked modalities such as RNA and protein. Additionally, scMODAL incorporates an extra regularizer to preserve the structures of the original datasets in the joint latent space, maintaining relative distances between cells. The benchmarking results on a collection of datasets in different scenarios highlight scMODAL's superiority in mixing cell distributions, maintaining cell-type separations, and accurately matching corresponding cell states across modalities.

Seurat_CCA/rPCA

Label transfer:

```
celltype.predictions <- TransferData(  
  anchorset = transfer.anchors, refdata = pbmc.rna$seurat_annotatons,  
  weight.reduction = pbmc.atac[["lsi"]], dims = 2:30) #Based on the labels of the  
nearest neighbors
```

Co embedding:

```
genes.use <- VariableFeatures(pbmc.rna)  
refdata <- GetAssayData(pbmc.rna, assay = "RNA", slot = "data")[genes.use, ]  
imputation <- TransferData(  
  anchorset = transfer.anchors, refdata = refdata,  
  weight.reduction = pbmc.atac[["lsi"]], dims = 2:30) #weighted sum of neighboring  
cells, the same features as in the reference
```

Seurat bridage

- Diagonal + joint-profiling

Seurat WNN

- Vertical
- WNN: weighted nearest neighbors, edges of different modality get different weights, sum of the weighted edges

Seurat bridge

Suppose:

- Reference: scRNA-seq (annotated cell types)
- Bridge: multiome (RNA + ATAC)
- Query: scATAC-seq

We want:

Transfer RNA cell labels to ATAC-only cells.

Steps:

1. Learn WNN embedding on the bridge dataset.
2. Map reference RNA into bridge.
3. Map ATAC query into bridge.
4. Compute anchors indirectly through bridge space.

That's what `FindBridgeTransferAnchors()` automates.

- **Reference dataset:** well-annotated, integrated, multi-modal or single-cell data
- **Query dataset:** new cells you want to classify or embed

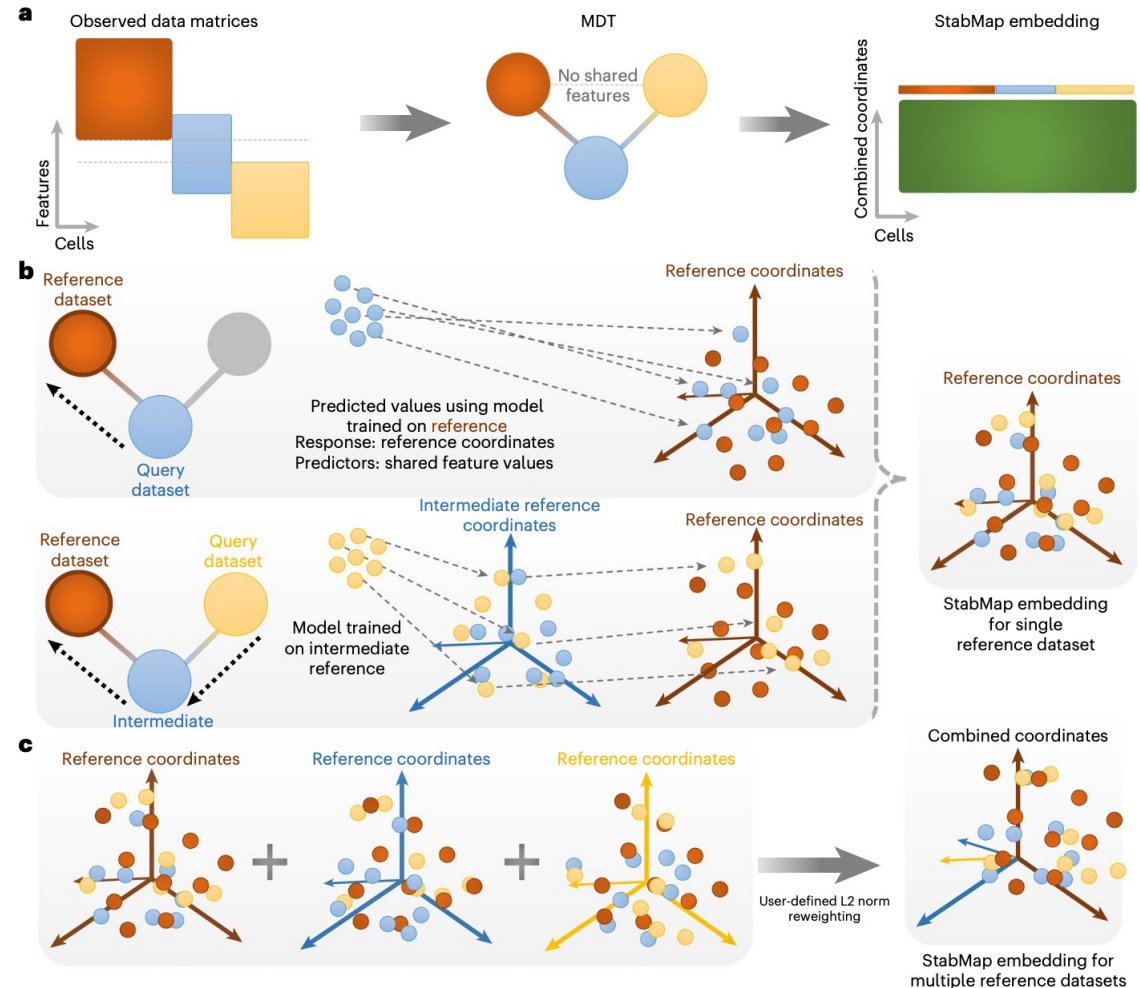
`MapQuery` allows you to:

1. Take the **anchors** (relationships) between reference and query
2. **Project the query** into the reference's low-dimensional space
3. **Predict metadata** (cell type, cluster, etc.) for the query
4. Optionally **impute missing modalities**

Linear algebra:
matrix multiplication
Data projection

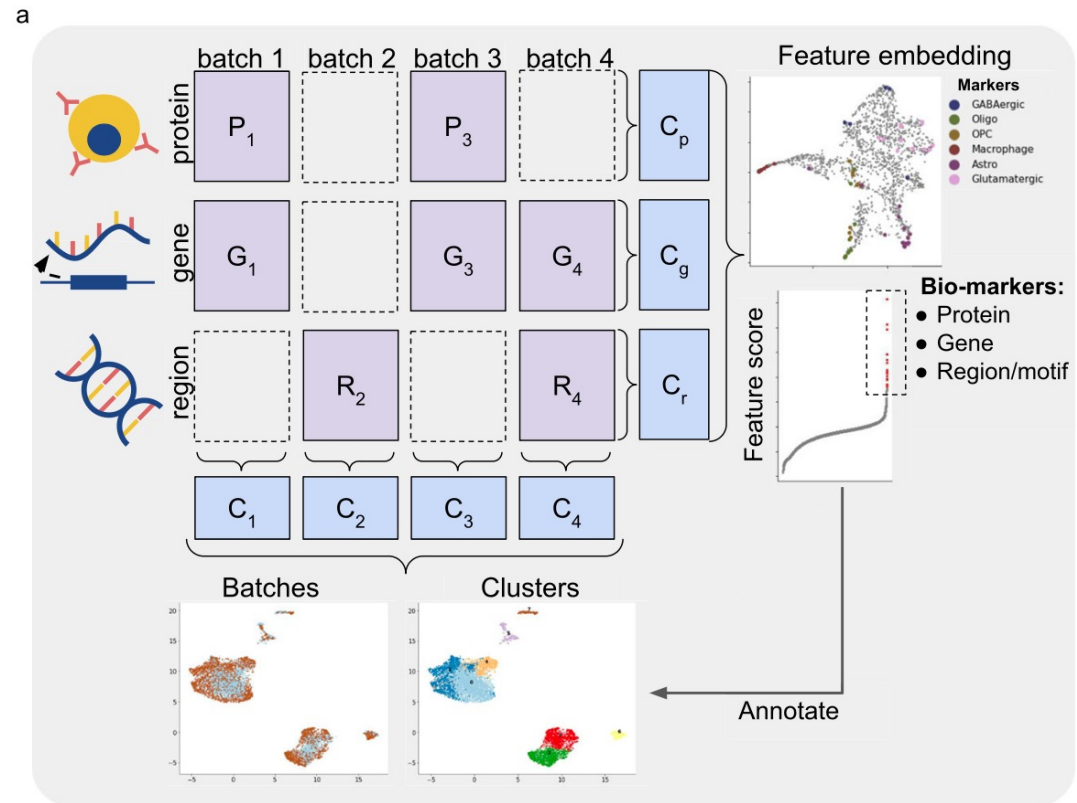
Stabmap

- Work on MDT: a connected network
- Shared features between some datasets
- Linear algebra (matrix multiplication)
- Project cells between low-dimensional spaces based on shared features (recalculate the feature loading using multivariate linear regression using only the shared features)
- Use both shared + nonshared features (dimensional reduction within datasets)
- One needs to remove batches using e.g. harmony afterwards



scMoMAT

Batches: share the same factors
Modality: share cells



b

$$\begin{matrix} e_j \\ \begin{matrix} \begin{matrix} e_i \\ X_{ij} \end{matrix} \end{matrix} \end{matrix} = \alpha_{ij} * \begin{matrix} \text{Factor } e_i \\ \begin{matrix} C_i \end{matrix} \end{matrix} * \begin{matrix} \text{Assoc} \\ \begin{matrix} \Sigma_{ij} \end{matrix} \end{matrix} * \begin{matrix} \text{Factor } e_j \\ \begin{matrix} C_j^T \end{matrix} \end{matrix} + \begin{matrix} \text{Bias } e_i \\ \begin{matrix} b_i \end{matrix} \end{matrix} + \frac{\text{Bias } e_j}{b_j}$$

Prior knowledge guidance: Mosaic integration

- Linked features (**partially share features — horizontal integration**)
 - ❖ RNA+ATAC (strong)
 - ❖ RNA+surfaceProtein (weak)
 - ❖ <https://www.nature.com/articles/s41467-021-25089-2>
 - Use only linked features:
 - E.g. **Seurat_CCA/rPCA**, **LIGER-iNMF**, **Portal**, ~~VIPCCA~~, ~~SCALEX~~, ~~iMAP~~
 - **Lose information**
 - Linked+unlinked features:
 - E.g. **scMODEL**, **GLUE**, **scVI**, **scANVI**, **MultiVI**, **LIGER-UINMF**, **MaxFuse**, **bindSC**, ~~Monae~~, ~~CoVEL~~
- Leverage joint-profiling (**partially share cells — vertical integration**)
 - E.g. **MIDAS**, **Seurat-bridge**, **Stabmap**, **MultiVI**, **Cobolt**, **Multigrade**, **scMoMAT**, **scVAEit**, ~~scButtery~~, and **SpatialGLUE**
- Leverage similar histology in spatial multi-omics
 - spatialEx+
- Pretrained foundation/generative models
 - Transfer learning, model refining: similarity between training data and query data

Linear:
NMF
Autoencoder

- iNMF:
 - Aligns gene programs
 - Not geometric coordinates
- PASTE2:
 - Aligns spatial + expression structure
- STalign:
 - Aligns geometry only